MTH203: Sequences and series
Lecture 07

1. Reviewing the notion of supremum

Recall that we have proved that if S is a set of real numbers, a real number x is the supremum of S if and only if it satisfies the following two conditions:

(a) x is an upper bound of S.
(b) Given any $\epsilon > 0$, there exists an element $s \in S$ such that $s > x - \epsilon$. Equivalently, $x - s < \epsilon$.

The inequality in (b) says that the *distance* between x and s is less than ϵ. (We will define the notion of *distance* formally soon.)

Note that the element s, whose existence is guaranteed in (b), *depends* on the ϵ that has been given to us. In other words, if a different positive real number ϵ' is given, the inequality $s > x - \epsilon'$ may not hold. So we may have to find *another* element s' of S such that $s' > x - \epsilon'$.

We will focus on this interpretation of condition (b) and use it to define *limits*.

2. Distance functions

Definition 1. Let S be a set. A *distance function* or a *metric* on S is a function $d : S \times S \to \mathbb{R}$ having the following properties:

(a) $d(x, y) \geqslant 0$ for all $x, y \in S$.
(b) $d(x, y) = 0$ if and only if $x = y$.
(c) $d(x, y) = d(y, x)$ for all $x, y \in S$.
(d) For any $x, y, z \in S$, we have $d(x, y) + d(y, z) \geqslant d(x, z)$.

The condition (d) in the above definition is called the *triangle inequality*. If you visualize x, y and z as vertices of a triangle and we interpret the distance to be the length of the sides joining the corresponding vertices, this reminds us of the familiar triangle inequality from Euclidean geometry. In fact, this definition is supposed to generalize our familiar notion of distance in the Euclidean plane.

Definition 2. A *metric space* is a set S equipped with a distance function.

Example 3. Let S be any set. We define a distance function on S by setting

$$d(x, y) = \begin{cases} 0 & \text{if } x = y, \\ 1 & \text{otherwise.} \end{cases}$$

It is very easy to verify that this really does satisfy all the conditions in the definition of a distance function.

Note that the distance function is *part of the notion of a metric space*. It is possible to define two different distance functions on the same set S. This gives us two different metric spaces (even though the underlying set may be the same). It would be more accurate to say that if d is a distance function on a set S, then the *ordered pair* (X, d) is a metric space. (In fact, this is the correct way to write it. I have deliberately chosen to be slightly informal in the above definition.)

3. The standard metric on the real numbers

In this course, we are not really concerned with arbitrary metric spaces. We would like to focus on the real numbers. First, let us recall the *absolute value* function. This is a function from $\mathbb{R}$ to itself, denoted by $x \mapsto |x|$ and defined by

$$|x| = \begin{cases} x & \text{if } x \geqslant 0, \\ -x & \text{otherwise.} \end{cases}$$

Let us prove a basic property of this function:

Lemma 4. *Let x by a real number. Then $|x| \geqslant x$ and $|x| \geqslant -x$.*

Proof. We will consider two cases: $x \geqslant 0$ and $x < 0$.

If $x \geqslant 0$, then $|x| = x$. Thus, $|x| \geqslant x$ as required. Since $x \geqslant 0$, we have $-x \leqslant 0$ and so $x \geqslant -x$. Thus, in this case $|x| \geqslant -x$. This proves the lemma in the case $x \geqslant 0$.

Now, suppose $x < 0$. Then $|x| = -x$. As $x < 0$, we have $-x > 0$. Thus, $-x > x$. Thus, $|x| > x$. Since $|x| = -x$, we trivially have $|x| \geqslant -x$. This proves the lemma in the case $x < 0$. $\square$

We will now define the *standard metric on* $\mathbb{R}$. This is the function $d : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ defined by $d(x, y) = |x - y|$. Let us verify that this really is a distance function. Conditions (a), (b) and (c) in Definition 1 are trivially verified. We will focus on condition (d).

Let x, y and z be real numbers. We would like to prove that

$$|x - z| \geqslant |x - y| + |y - z|.$$

By Lemma 4, we have the inequalities $|x-y| \geqslant x-y$ and $|y-z| \geqslant y-z$. Adding these two inequalities together, we get

$$|x - y| + |y - z| \geqslant x - y + y - z = x - z.$$

Similarly, by Lemma 4, we have $|x - y| \geqslant -(x - y)$ and $|y - z| \geqslant -(y - z)$. Adding these inequalities together, we get

$$|x - y| + |y - z| \geqslant -(x - y) + (-(y - z)) = -x + y - y + z = -(x - z).$$

We conclude that $|x-y|+|y-z| \geqslant |x-z$ since $|x-z|$ is equal to either $(x-z)$ or $-(x - z)$. This proves that condition (d) in Definition 1 holds. Thus, the function we have defined is actually a distance function on $\mathbb{R}$. We will call this the *standard metric on* $\mathbb{R}$.

4. Neighbourhoods

Definition 5. Let S be a metric space with distance function d. Let ϵ be any positive real number and let x be an element of S. Then, the *ϵ-neighbourhood of x* in S is the set

$$V_\epsilon(x) := \{y \in S | d(x, y) < \epsilon\}$$

Thus, the ϵ-neighbourhood of x is the set of all points in S which are at distance less than ϵ. You may visualize this as a ball of radius ϵ with centre at x. The points of the neighbourhood are the points that are in the "interior" of the ball. The points on the "boundary", i.e. the points that are at distance ϵ from x, are not included in the ϵ-neighbourhood.

Let us see what this means for the standard metric on $\mathbb{R}$.

Example 6. We consider $\mathbb{R}$ as a metric space with the standard metric. Let x be a real number and let $\epsilon > 0$. We claim that

$$V_\epsilon(x) = \{y \in \mathbb{R} | x - \epsilon < y \text{ and } y < x + \epsilon\}.$$

In other words, we claim that $V_\epsilon(x)$ is just the *open interval* $(x - \epsilon, x + \epsilon)$.

First, let us prove that $V_\epsilon(x) \subset (x - \epsilon, x + \epsilon)$. Suppose y is an element of $V_\epsilon(x)$. Then $|x - y| < \epsilon$. We consider two cases: $x \leqslant y$ and $x > y$.

Suppose $x \leqslant y$. Then $x - y \leqslant 0$ and so $|x - y| = -(x - y) = -x + y$. Since $y \in V_\epsilon(x)$, we have $-x + y < \epsilon$. Adding x to both sides of this inequality, we get $x - x + y < x + \epsilon$. Thus, $x + \epsilon > y$. On the other hand, since $\epsilon > 0$, $x - \epsilon < x$. So, the inequalities $x \leqslant y$ and $x - \epsilon < x$ give us the inequality $x - \epsilon < y$. Thus, we see that $y \in (x - \epsilon, x + \epsilon)$ in the case $x \leqslant y$.

Now, suppose $x > y$. Then $x - y > 0$ and so $|x - y| = x - y$. Thus, $x - y < \epsilon$. Adding $y - \epsilon$ to both sides of this inequality, we get

$$x - y + y - \epsilon < \epsilon + y - \epsilon.$$

Thus, $x - \epsilon < y$. On the other hand, since $y - x < 0$ and $\epsilon > 0$, we have $y - x < \epsilon$. Adding x to both sides of this inequality, we get $y < x + \epsilon$. Thus, we have proved that $y \in (x - \epsilon, x + \epsilon)$ in the case $x > y$.

We conclude that $V_\epsilon(x) \subset (x - \epsilon, x + \epsilon)$. Now, let us prove that $(x - \epsilon, x + \epsilon) \subset V_\epsilon(x)$.

If $y \in (x - \epsilon, x + \epsilon)$ we have the two inequalites $x - \epsilon < y$ and $y < x + \epsilon$. Adding $\epsilon - y$ to both sides of the first inequality gives us $x - y < \epsilon$, while adding $-x$ to both sides of the second inequality gives us $y - x < \epsilon$. Thus ϵ is greater than both $(x - y)$ and $-(x - y)$ and hence it is greater than $|x - y$. This is exactly what we wanted to prove.

Thus, we have proved that $V_\epsilon(x) = (x - \epsilon, x + \epsilon)$.

Note that there is a relationship between ϵ-neighbourhoods for various values of ϵ. A very obvious fact is as follows:

$$\epsilon_1 > \epsilon_2 \implies V_{\epsilon_2} \subset V_{\epsilon_1}$$

Thus, as ϵ becomes smaller and smaller, the ϵ-neighbourhood $V_\epsilon(x)$ also becomes smaller and smaller.

Lemma 7. *Let S be a metric space and let x be a point of S. Then, we have*

$$\bigcap_{\epsilon > 0} V_\epsilon(x) = \{x\}.$$

Proof. For any $\epsilon > 0$, we have $d(x, x) = 0 < \epsilon$ and so $x \in V_\epsilon(x)$. Thus, $x \in \bigcap_{\epsilon > 0} V_\epsilon(x)$.

On the other hand, if $y \in \bigcap_{\epsilon > 0} V_\epsilon(x)$, then $d(x, y) < \epsilon$ for any $\epsilon > 0$. Thus, $d(x, y)$ cannot be a positive number. (This is because if it were a positive number, we would be able to choose another positive number smaller than it and call it ϵ. For example, if $d(x, y) > 0$, then $0 < d(x, y)/2 < d(x, y)$.) Thus, $d(x, y) = 0$, which implies that $x = y$. ☐

5. Limit of a sequence

Recall that a sequence in any set S is a function from $\mathbb{N}$ to S. We will now consider sequences in a metric space.

Definition 8. Let S be a metric space. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in S. Let x be an element of S. We say that the sequence $(x_n)_{n \in \mathbb{N}}$ converges to x if, given $\epsilon > 0$, one can find a natural number N_ϵ (depending on ϵ) such that for any $n > N_\epsilon$, we have $x_n \in V_\epsilon(x)$.

Suppose the sequence is written as follows:

$$x_1, x_2, \ldots, x_{N_\epsilon - 1}, x_{N_\epsilon}, x_{N_\epsilon + 1}, \ldots$$

Then, we are saying that all the points of x_{N_ϵ} are in $V_\epsilon(x)$. Thus the "tail" of the sequence, beyond *some* point lies in the ϵ-neighbourhood. If ϵ is taken to be very small, N_ϵ will have to be taken very large. However, the point is that such an N_ϵ can always be found, if x is the limit of the sequence.

Let us rewrite this definition specifically for sequences of real numbers.

Definition 9. Let $(x_n)_{n \in \mathbb{N}}$ be a sequence in $\mathbb{R}$. Let x be an element of S. We say that the sequence $(x_n)_{n \in \mathbb{N}}$ converges to x if, given $\epsilon > 0$, one can find a natural number N_ϵ (depending on ϵ) such that for any $n > N_\epsilon$, we have $|x - x_n| < \epsilon$, or equivalently, $x_n \in (x - \epsilon, x + \epsilon)$.

We set up a little more terminology:

Definition 10. Let S be a metric space and let $(x_n)_{n\in\mathbb{N}}$ be a sequence in S.

(a) We say that the sequence $(x_n)_{n\in\mathbb{N}}$ *converges* if it converges to some $x \in S$.
(b) We say that the sequence $(x_n)_{n\in\mathbb{N}}$ diverges if it does not converge.
(c) If the sequence $(x_n)_{n\in\mathbb{N}}$ converges to x, we say that x is the *limit* of the sequence $(x_n)_{n\in\mathbb{N}}$. The limit of a sequence $(x_n)_{n\in\mathbb{N}}$ is denoted by $\lim_{n\to\infty} x_n$. (We may also write it as $\lim_n x_n$ or $\lim x_n$ if there is no fear of confusion.)

Example 11. Let $(x_n)_{n\in\mathbb{N}}$ be the sequence of real numbers, defined by $x_n = 1/n$. Then, we claim that $\lim x_n = 0$. Indeed, we know that given any $\epsilon > 0$, there exists an integer N such that $1/N < \epsilon$. If $n > N$, then $1/n < 1/N$ and hence $1/n < \epsilon$. For any such n, we have $|1/n - 0| = 1/n < \epsilon$. Thus, $\lim x_n = 0$.

Example 12. Let $(x_n)_{n\in\mathbb{N}}$ be the sequence of real numbers, defined by $x_n = (-1)^n$. The first few terms of this sequence can be written as follows:

$$-1, 1, -1, 1, -1, 1, \ldots$$

In other words, $x_n = -1$ if n is odd and $x_n = 1$ if n is even.

We claim that $(x_n)_n$ diverges. Indeed, suppose this sequence converges to some point $x \in \mathbb{R}$. We will show that there exists some ϵ for which there does not exist any natural number N such that $x_n \in V_\epsilon(x)$ for every $n > N$.

Indeed, let us take $\epsilon = 1$. Suppose that there exists some N such that for any $n > N$, we have $|x - x_n| < 1$. Take n to be any odd number greater than N. Then $x_n = -1$ and so $|x - (-1)| < 1$. Since $n + 1$ is even, we have $x_{n+1} = 1$. Since $n + 1 > n > N$, we must have $|x - 1| < 1$. Then, the triangle inequality implies

$$1 + 1 > |x - (-1)| + |x - 1| = |(-1) - x| + |x - 1| \geqslant |(-1) - 1| = |-2| = 2.$$

This is a contradiction. Thus, we see that the sequence $(x_n)_n$ diverges.

It is very important to understand this example clearly. Note that 1 occurs infinitely many times in this sequence. Thus, $|1 - x_n| = 0$ for infinitely many values of n. However, for any $N > 0$, if we take m to be an *odd* integer greater than N, then $|1 - x_m| = 2$. Thus, no “tail” of this sequence can end up in a sufficiently small neighbourhood of 1. For a similar reason, -1 also cannot be the limit of this sequence even though it appears in the sequence infinitely many times.

6. A question

Let S be a metric space and let $(x_n)_n$ be a sequence in S. Suppose that this sequence converges. This means that it converges to *some* element x of S. However, is this element unique? In other words, is it possible that there is some *other* element y such that the sequence converges to y as well?

We will see in the next lecture that this cannot happen. The limit of a sequence, *if it exists*, must be unique.

7. Reading suggestions

This lecture corresponds to Section 2.2 of the textbook ([1]). The book does not talk about metric spaces in detail in this section. A more detailed discussion of metric spaces is given in Section 8.2 (which is not part of the syllabus for this course).

We will not need metric spaces for this course. We have looked at the notion of metric spaces simply to give a better perspective of the subject matter.

References

[1] Stephen Abbott, *Understanding Analysis*, Undergraduate Texts in Mathematics, Springer, 2015. 4